

Simulation Plan

Oren Ram
Ziv Abraham

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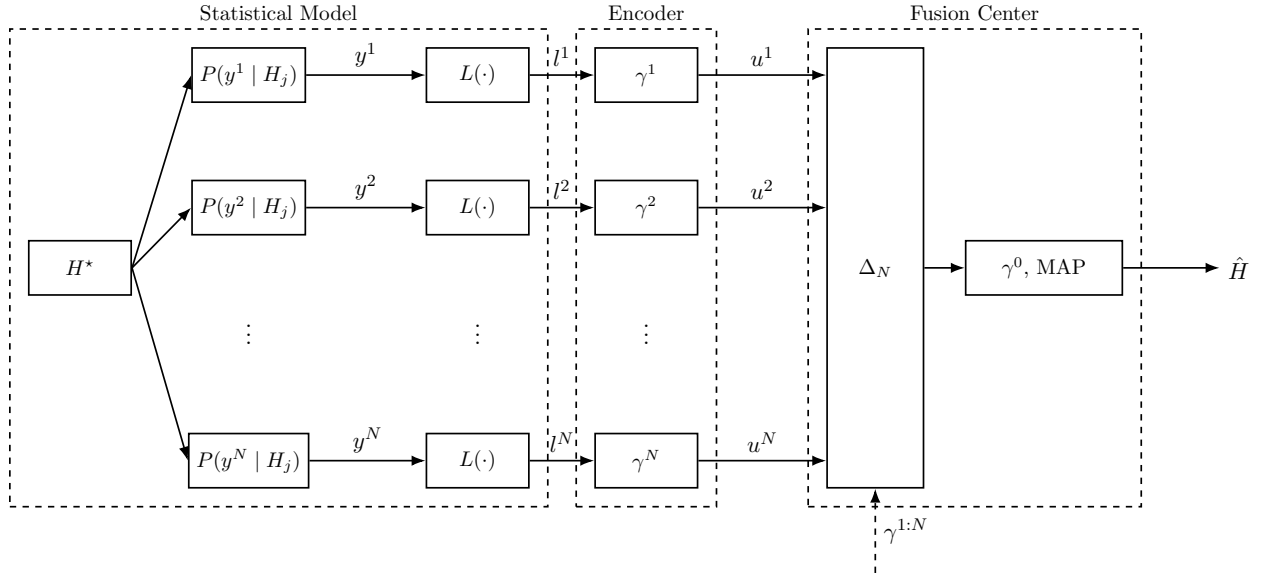


Figure 1: Block diagram of the decentralized detection system. The source H^* generates the hypothesis observed by N sensors as $y^i | H_j$. Each sensor reduces its observation to the likelihood ratio $l^i = L(y^i)$, which is a sufficient statistic for H , and quantizes it via the threshold policy γ^i into a message u^i . The fusion center forms the statistic Δ_N from all messages together with the realized encoding policies $\gamma^{1:N}$ (dashed), and applies the MAP rule γ^0 to produce the decision $\hat{H}$.

Simulation of Silence is free (SIF) VS Vanilla setups with $|\mathcal{U}| = 4$

We would like to compare the Error exponents achieved for different setups under the same total rate budget. For that we should define the **Expected Rate per Sensor**:

$$r(\gamma) \triangleq \mathbb{E}[\text{bits sent}] = \sum_{u \in \mathcal{U}} P^\gamma(u) \rho(u) \quad (1)$$

$$P^\gamma(u) = pP^\gamma(u|H_1) + (1-p)P^\gamma(u|H_2) \quad (2)$$

The two schemes will result in the following $r(\gamma)$:

- **Vanilla**: Every sensor always sends its action, so the average bits used is $\log_2(|\mathcal{U}|)$
- **SIF**: Sends the action only if $u \notin \mathcal{S}$, when $\mathcal{S}$ is the group of silent actions. In this case, every symbol costs $\log_2(|\mathcal{U}|)$ but is sent w.p $\delta = P(u \notin \mathcal{S})$, so $r_{SIF} = \delta \log_2(|\mathcal{U}|)$.